UNIT-5 COORDINATION CHEMISTRY

MCQ:

1. In the complex $\left[M(en)_2(Ox)\right]Cl$ For the central metal ion M^{3+}

   The primary valance is = +3

   The secondary valance = 6

   sum of primary valance and secondary valance = 3+6 = 9

   Answer : option (d)

2. The complex is $\left[M(H_2O)_5 Cl\right]Cl_2$

   1000 ml of 1M solution of the complex gives 2 moles of Cl^- ions

   1000 ml of 0.01M solution of the complex will give

   $$\frac{100 \text{ ml} \quad 0.01M \quad 2Cl^-}{1000 \text{ ml} \quad 1M} = 0.002 \text{ moles of } Cl^- ions$$

   Answer : option (b)

3. Molecular formula: $MSO_4Cl.\,6H_2O$.

   Formation of white precipitate with Barium chloride indicates that SO_4^{2-} ions are outside the coordination sphere, and no precipitate with $AgNO_3$ solution indicates that the Cl^- ions are inside the coordination sphere.Since the coordination number of M is 6, Cl^- and 5 H_2O are ligands, remaining 1 H_2O molecular and SO_4^{2-} are in the outer coordination sphere.

   Answer : option (c)

4. $\left[\overset{+}{Fe}(H_2O)_5\overset{+}{NO}\right]^{2+} SO_4^{2-}$

   +1 and +1 respectively

   Answer : option(d)

5. Answer : option(d)

6. Answer : option(d)

7. Answer : option(c)

   $Ti^{4+}\left(d^0 \Rightarrow 0BM\right)$

   $Co^{2+}\left(d^7 \text{spin free} \Rightarrow t_{2g}^{\ 5}, e_g^{\ 2}\ ;\ n=3\ ;\ \mu = 3.9BM\right)$

   $Cu^{2+}\left(d^9 \text{Low spin} \Rightarrow t_{2g}^{\ 6}, e_g^{\ 3}\ ;\ n=1\ ;\ \mu = 1.732BM\right)$

   $Ni^{2+}\left(d^8 \text{Low spin} \Rightarrow t_{2g}^{\ 6}, e_g^{\ 2}\ ;\ n=2\ ;\ \mu = 2.44BM\right)$

8. Answer : option(b)

   The electronic configuration $t_{2g}^{\ 3}, e_g^{\ 2}$

   $\left[3\times(-0.4)+2(0.6)\right]\Delta_0$

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